

SQL Analysis Summary Report

Project : E-Commerce Revenue & Growth Intelligence Platform

1. SQL Analysis Overview

SQL analysis was performed on the cleaned Brazilian E-Commerce (Olist) dataset to understand the business performance, customer behavior, sales activity, revenue trends, product performance, seller performance, payment behavior, customer reviews, and geographic distribution.

The cleaned datasets were imported into a MySQL database named **olist_ecommerce**. SQL queries were then created and organized into separate analysis files.

The SQL analysis focused on converting the cleaned data into meaningful business metrics and insights that can later be represented through Power BI dashboards.

2. SQL Analysis Files

The SQL analysis was organized into the following files:

File	Analysis Area
01_Basic_Counts.sql	Basic record counts
02_Basic_Aggregations.sql	Basic revenue and price calculations
03_Order_Analysis.sql	Order status, dates, volume and order behavior
04_Revenue_Analysis.sql	Revenue, freight, categories and yearly/monthly revenue
05_Product_Analysis.sql	Product categories, prices, weight and dimensions
06_Customer_Analysis.sql	Customer distribution and repeat-customer analysis
07_Payment_Analysis.sql	Payment methods, values and installments
08_Review_Analysis.sql	Review scores and customer satisfaction
09_Seller_Analysis.sql	Seller performance and revenue
10_Geographic_Analysis.sql	Customer, seller and geographic analysis
11_KPI_Queries.sql	Key Performance Indicators
12_Monthly_Growth.sql	Yearly/monthly growth analysis
13_Business_Insights.sql	Business-focused analysis and interpretation

✓ **Total SQL files completed: 13**

3. Main Business KPIs

The SQL analysis produced the following important KPIs:

KPI	Result
Total Customers	99,441
Total Orders	99,441
Total Products	32,951
Total Sellers	3,095
Total Items Sold	112,650
Product Revenue	R\$13,591,643.70
Total Freight	R\$2,251,909.54
Sales Value Including Freight	R\$15,843,553.24
Average Order Value	R\$137.75
Average Item Price	R\$120.65
Delivered Order Rate	97.02%
Cancellation Rate	0.63%
Positive Review Rate	77.07%
Average Review Score	4.09 / 5
Average Items per Order	1.14
Total Payment Value	R\$16,008,872.12
Average Payment Value	R\$154.10
Average Payment Installments	2.85
Geolocation Coverage	99.72%

4. Order Analysis

The order analysis showed that the majority of orders were successfully delivered.

- Delivered orders: **96,478**
- Delivered order rate: **97.02%**
- Canceled orders: **625**
- Cancellation rate: **0.63%**
- Non-delivered orders: **2,963**
- Average items per order: **1.14**
- Average order value: **R\$137.75**

The highest number of orders occurred in **August**, with **10,843 orders** when all available years are combined.

The data also showed that Monday had the highest order volume among weekdays.

5. Revenue Analysis

Total product revenue generated from order items was:

R\$13,591,643.70

Total freight was:

R\$2,251,909.54

Therefore, product revenue plus freight resulted in:

R\$15,843,553.24

Revenue increased significantly between 2017 and the available 2018 period.

Year Revenue

2016 R\$49,785.92

2017 R\$6,155,806.98

2018 R\$7,386,050.80

Revenue growth from 2017 to 2018 was approximately **19.99%**.

The very high 2016–2017 growth percentage should not be interpreted as normal annual business growth because the dataset contains only a limited period of 2016 data.

6. Monthly and Growth Analysis

The monthly growth analysis showed several important trends.

Highest Revenue Month

November 2017

Revenue:

R\$1,010,271.37

Highest Order Volume Month

November 2017

Orders:

7,544

2017 → 2018 Growth

- Revenue growth: **19.99%**
- Order growth: **19.76%**

Average order value changed from:

R\$138.09 → R\$137.35

This indicates that revenue growth was mainly driven by **increased order volume**, rather than a significant increase in the amount spent per order.

The unusually large monthly growth percentages involving early 2017 are influenced by the very small number of records in the corresponding 2016 months.

7. Product Analysis

Product analysis was performed using product information and order-item data.

The highest-revenue product category was:

beleza_saude — R\$1,258,681.34

Other major revenue-generating categories included:

- relorios_presentes
- cama_mesa_banho
- esporte_lazer
- informatica_acessorios

The top 10 product categories together contributed approximately **62.36% of total product revenue**.

This indicates that a relatively small group of product categories contributes a large portion of overall revenue.

8. Customer Analysis

Customer analysis examined customer distribution, customer states, cities, order frequency, and repeat-customer behavior.

The largest customer base was in:

São Paulo (SP) — 41,746 customers

São Paulo also generated the highest customer-state revenue:

R\$5,202,955.05

The customer analysis also showed that the standard `customer_id` is associated with one order in this dataset. Therefore, `customer_unique_id` is more appropriate when analyzing true repeat-customer behavior.

9. Payment Analysis

Payment analysis showed that **credit card** was the dominant payment method.

- Credit card payment records: **76,795**
- Credit card share: **73.92%**
- Credit card average payment: **R\$163.32**
- Average number of installments: **2.85**

The total payment value recorded in the payment table was:

R\$16,008,872.12

Payment value is kept as a separate metric from product revenue because the payment table can contain multiple payment records for an order.

10. Review and Customer Satisfaction Analysis

The review analysis showed generally positive customer feedback.

- Total review records: **99,224**
- Average review score: **4.09 / 5**
- Positive reviews: **76,470**
- Positive review rate: **77.07%**
- Negative reviews: **14,575**
- Negative review rate: **14.69%**

Delivered orders had an average review score of:

4.16 / 5

Several non-delivered or intermediate order statuses had substantially lower average review scores.

This suggests an association between successful order completion and better customer satisfaction.

11. Seller Analysis

Seller analysis showed that all **3,095 sellers** had orders.

However, seller performance was highly uneven.

Seller Revenue Distribution

Revenue Range	Sellers
Below R\$1,000	1,667
R\$1,000–R\$4,999	862
R\$5,000–R\$9,999	274
R\$10,000–R\$49,999	252
R\$50,000+	40

The highest-revenue seller generated:

R\$229,472.63

No single seller dominated the marketplace; the top seller contributed approximately **1.69%** of total product revenue.

This indicates a broad seller base with significant differences in seller performance.

12. Geographic Analysis

Geographic analysis examined customer and seller distribution across Brazilian states and cities.

São Paulo was the dominant customer market:

- Customers: **41,746**
- Revenue: **R\$5,202,955.05**

The cleaned geolocation dataset contained:

738,332 records

Customer geolocation coverage was:

99.72%

- Customers with geolocation: **99,163**
- Customers without geolocation: **278**

This provides strong geographic coverage for future visualization and geographic analysis in Power BI.

13. Key Business Insights

The SQL analysis produced the following major business insights:

Insight 1 — Revenue Growth

Revenue increased by approximately **19.99% from 2017 to 2018**.

Insight 2 — Order Volume Drives Growth

Orders increased by approximately **19.76%**, while average order value slightly decreased. Therefore, growth was primarily driven by **higher order volume**.

Insight 3 — Revenue Concentration

The top 10 product categories generated approximately **62.36% of total product revenue**.

Insight 4 — Strongest Product Category

beleza_saude was the highest-revenue category with approximately **R\$1.26M** in product revenue.

Insight 5 — Strong Customer Market

São Paulo was the largest customer market and generated approximately **R\$5.20M** in revenue.

Insight 6 — Seller Performance Variation

Most sellers generated relatively low revenue, while a small group generated high revenue. This indicates significant variation in seller performance.

Insight 7 — Payment Preference

Credit cards were the dominant payment method, representing approximately **73.92%** of payment records.

Insight 8 — Customer Satisfaction

The average review score was **4.09**, with **77.07% positive reviews**.

Insight 9 — Delivery and Satisfaction

Delivered orders had a higher average review score than several non-delivered/intermediate statuses, suggesting that successful order completion is associated with better customer experience.

Insight 10 — Geographic Data Quality

Customer geolocation coverage reached **99.72%**, making the geographic data suitable for dashboard analysis.

14. SQL Analysis Conclusion

The SQL analysis transformed the cleaned e-commerce data into structured business information covering **sales, revenue, products, customers, sellers, payments, reviews, geography, KPIs, and growth**.

The analysis shows that the business experienced strong growth in the available 2018 period, with revenue and order volume both increasing by approximately 20% compared with 2017. Revenue is concentrated among major product categories, São Paulo is the strongest customer market, credit cards dominate payment behavior, and customer satisfaction is generally positive.

These SQL results provide the foundation for the next stage of the project: **Power BI dashboard development and business representation**.

The most important SQL-derived insights will be converted into **KPIs, charts, filters, dashboard visuals, business recommendations, and Review 2 presentation points**.

15 Business Insights

#	Business Insight	Key Result
1	Highest Revenue Year	2018 — R\$7,386,050.80
2	Revenue Growth	2017 → 2018: 19.99%
3	Order Growth	2017 → 2018: 19.76%
4	Main Growth Driver	Growth was mainly driven by higher order volume , while AOV slightly decreased
5	Highest Revenue Month	November 2017 — R\$1,010,271.37
6	Top Product Category	beleza_saude — R\$1,258,681.34
7	Revenue Concentration	Top 10 categories contributed 62.36% of product revenue
8	Top Customer State by Revenue	São Paulo (SP) — R\$5,202,955.05
9	Top Seller by Revenue	Seller 4869f7...52b2 — R\$229,472.63
10	Seller Revenue Distribution	1,667 sellers generated below R\$1,000; only 40 generated R\$50,000+
11	Dominant Payment Method	Credit Card — 73.92% of payment records
12	Customer Satisfaction	77.07% positive reviews; average score 4.09/5
13	Delivery & Customer Satisfaction	Delivered orders had 4.16/5 average review score, higher than several other statuses
14	Geographic Data Coverage	99.72% of customers have geolocation coverage
15	Overall Business Performance	99,441 customers, 99,441 orders, R\$13.59M product revenue, 97.02% delivered rate, 4.09 average review